AWS Cloud9 for Python
Cloud Application Development with AWS SDK
Tutorial

Topics

1. Setup AWS Cloud9 for Python (section 1)
2. Python – Working with virtual environments (section 2)

You MUST ensure that

- **!!!You use the AWS resources responsibly!!!**

  To prevent ongoing charges to the AWS account after you're done with this tutorial, **stop** the EC2 instance that hosts your AWS Cloud9 environment.

- When **naming** different cloud resources, you **must prefix** all the cloud resources you used **with your student ID**.

1 Setting up Cloud9 for Cloud Application Development in Python

- Create a Cloud9 environment

  1. From the *AWS Management Console* search and select *Cloud9* (see Figure 1 for an example)
  2. Create an AWS Cloud9 Environment by clicking on the *Create Environment* button (see Figure 2 for an example)
  3. *Step 1 Name environment*: in the *Name* text box provide a name for your AWS Cloud9 environment, and optionally a description, and then press the *Next* button (see Figure 3 for an example)

4. *Step 2 Configure Settings*: for the *Environment type* select *EC2 instance for environment*, for the *Instance type* select *t2.micro*, and for *Platform* select *Amazon Linux 2* (see Figure 4 for an example)
5. *Step 3 Review* the details of your Cloud9 environment, if you are happy with the settings click on *Create environment* button to create your Cloud9 environment (see Figure 5 and Figure 6 for an example).

- Follow *steps 1* to *4* inclusive (except the item named "Set Up Credentials in Your Environment") from the *Python Tutorial for AWS Cloud9*, to set up your Cloud9 development environment to work with AWS SDK for Python (Boto3) https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html
  - ***Note: the name of the S3 bucket must be unique across all regions!***

!!!Use the AWS resources responsibly!!!

To prevent ongoing charges to the AWS account after you're done with this tutorial, **stop** the EC2 instance that hosts your AWS Cloud9 environment.

2 Python – Working with virtual environments

Virtual environments allow us to manage separate package installations for different projects. For more information please consult https://packaging.python.org/guides/installing-using-pip-and-virtual-environments/

1. Create a virtual environment: to create a virtual environment, navigate (by using the command *cd*) to your project's directory and run the command *venv* at the terminal. The following command will create the virtual environment named env.

   ```
   $ python3 -m venv env
   ```

2. Activate a virtual environment: before we can start installing or using packages in the virtual environment we have to activate the virtual environment we created, by running the following command at the terminal

   ```
   $ source env/bin/activate
   ```

   Note that once the virtual environment is activated the terminal's prompt is prefixed by the name of your virtual environment (i.e. (env) in this example).

3. Deactivate the virtual environment: when you want to exit the virtual environment (for example when you finish a working session, or would like to work on another project) run the following command at the terminal

   ```
   $ deactivate
   ```

4. Each time you want to enter again the virtual environment for your project follow step 2. Note that typically we run only once *item 1* per project, and as many times needed *item 2* and *item 3*.

Freezing dependencies

We can export a list off all installed packages and their versions by running the command *pip freeze* at the terminal

```
$  pip freeze
```

Resources

- AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/welcome.html
- Python Tutorial for AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html
- Amazon S3 https://aws.amazon.com/s3/
- Python Documentation https://docs.python.org/
- Python Tutorial https://docs.python.org/3/tutorial/index.html

Appendix A - Creating a Cloud9 environment

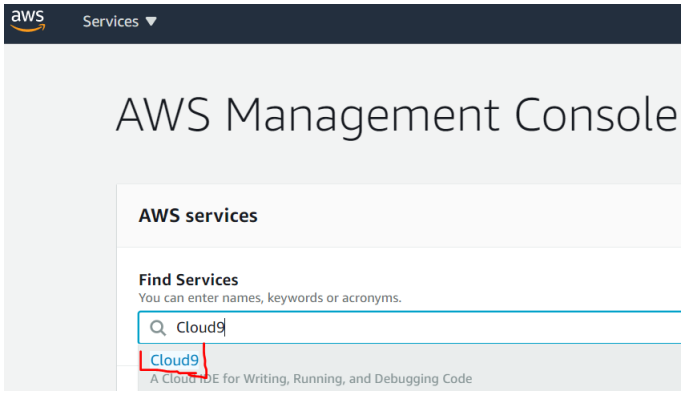


Figure 1: How to Search for Cloud9 via the AWS Management Console

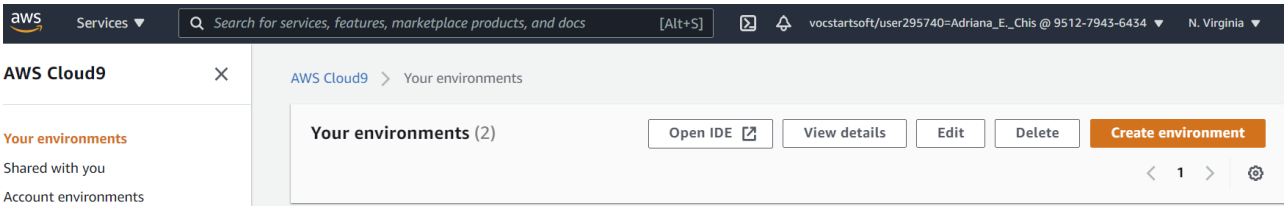


Figure 2: How to Create a Cloud9 Environment

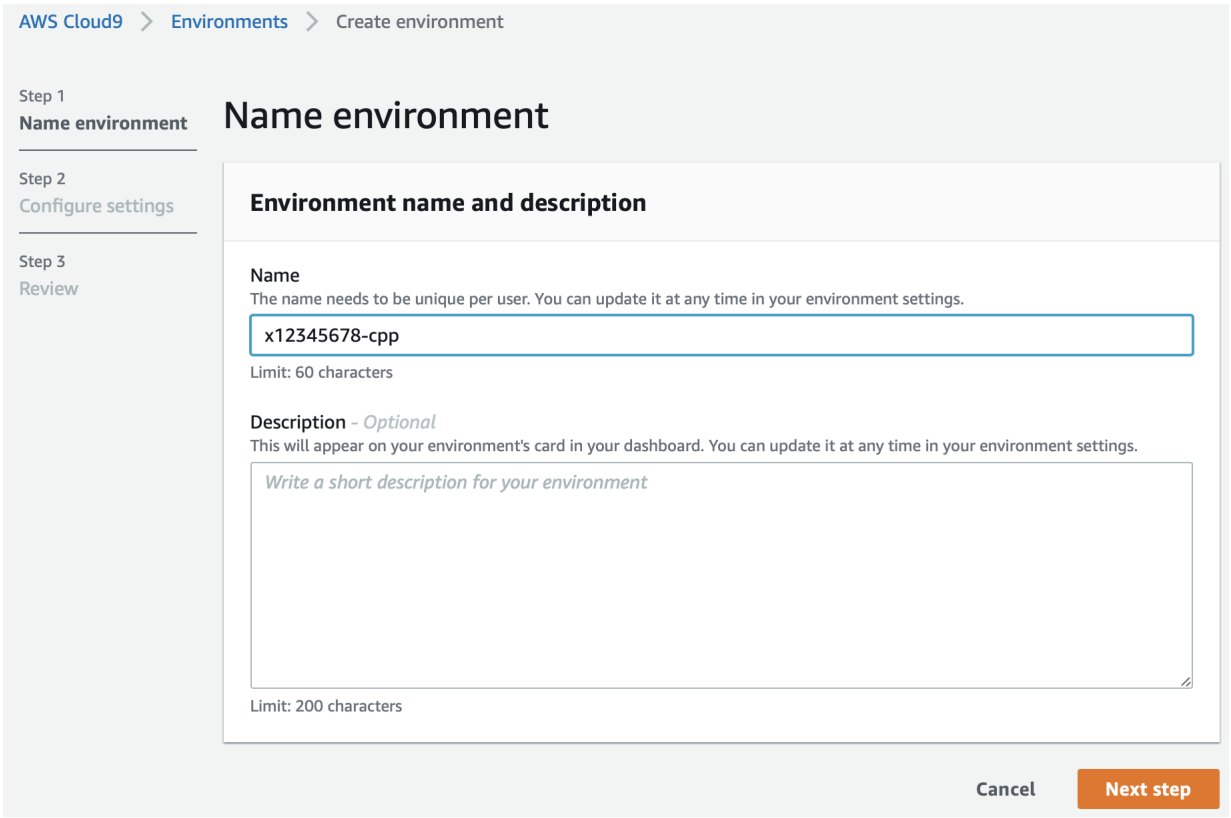


Figure 3: How to Create a Cloud9 Environment

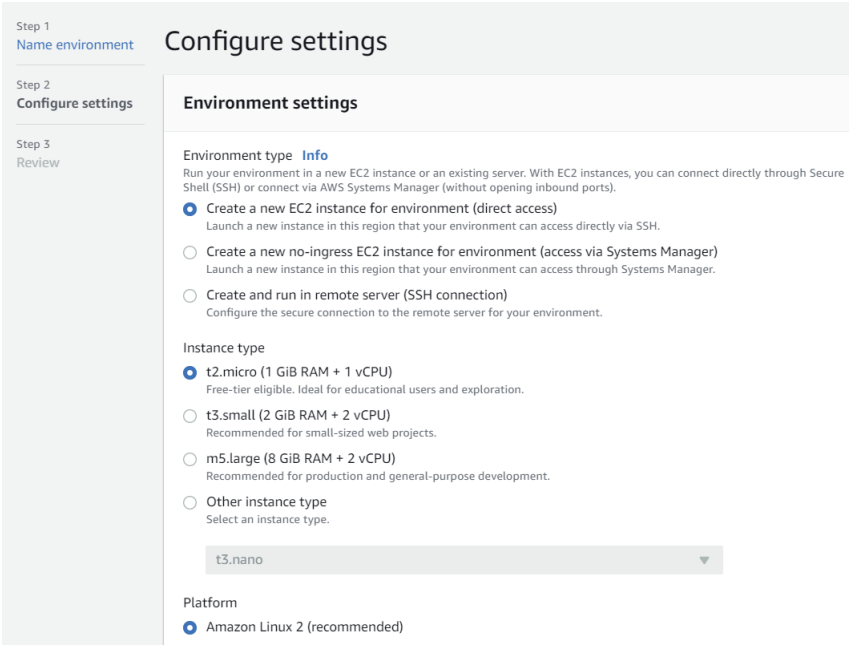


Figure 4: How to Create a Cloud9 Environment

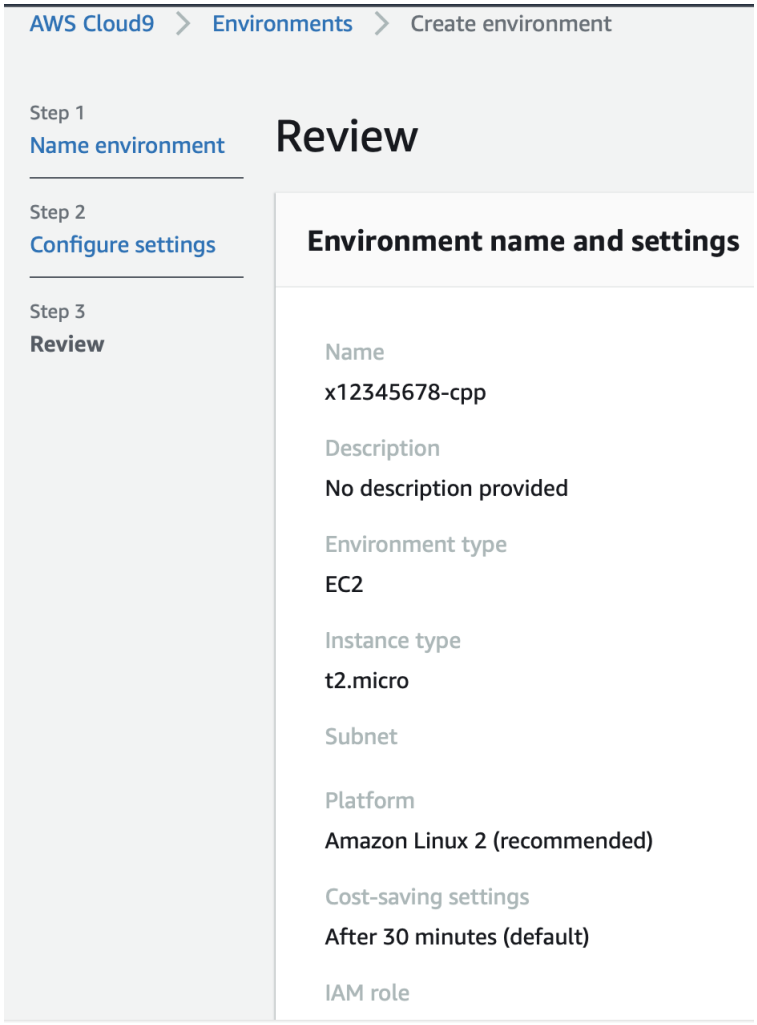


Figure 5: How to Create a Cloud9 Environment

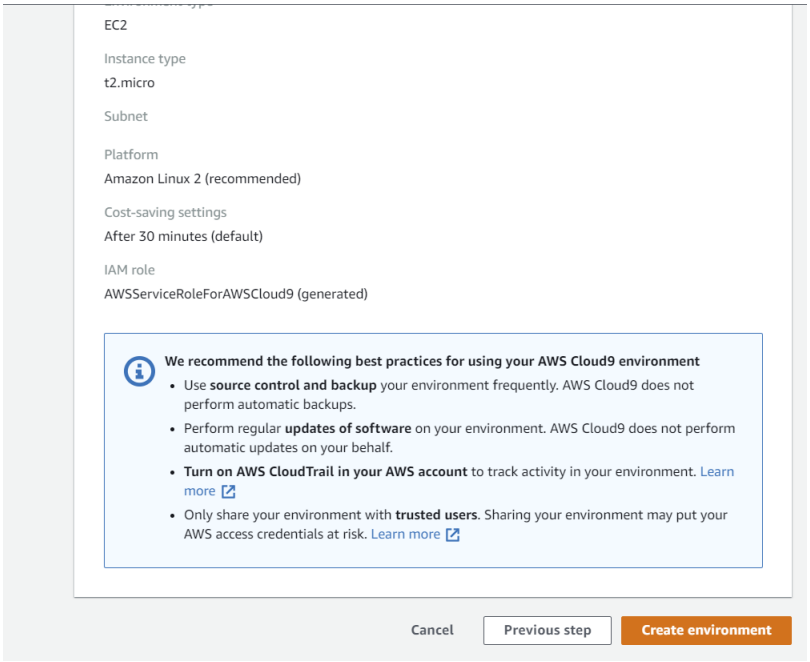


Figure 6: How to Create a Cloud9 Environment